DVA Project Report

Analysis of Reels Consumption Across Different Social Media Platform

Field	Details
Sector	Digital Health & Consumer Technology
Team ID	G - 2
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Research Question	How does short-form content consumption impact attention span across user segments?

2. Executive Summary

Short-form video platforms — TikTok, Instagram Reels, and YouTube Shorts — have fundamentally reshaped digital consumption habits over the last five years. This project investigates whether and how daily exposure to such content correlates with measurable changes in attention span, focus, and cognitive productivity across different demographic and behavioral segments.

8,926	3.0 hrs	48.2%	19.3%
Total Users Analysed	Avg Daily Reels	Users Over-Consuming	Danger Zone Users

Problem

Nearly half of surveyed users consume more than three hours of short-form video content daily, against a backdrop of only 6.5 hours of average sleep — 1.5 hours below medical recommendations. The core question is whether these consumption patterns causally degrade cognitive performance metrics, and which user segments face the greatest risk.

Approach

An 8,926-record dataset covering behavioural and cognitive variables was subjected to a full data cleaning pipeline followed by structured EDA encompassing univariate, bivariate, and multivariate analysis. Feature engineering produced a Cognitive Performance Index (CPI), consumption tiers, and a four-category risk segmentation model. Four Tableau dashboards were designed to operationalise the findings.

Key Insights

- No single variable is a reliable predictor of attention decline — the relationship is multi-factorial.
- Young adults aged 19–24 are the highest-risk group: lowest mean attention score (5.38) and highest average consumption (3.1 h/day).
- 19.3% of users simultaneously exceed 3 h reels AND sleep under 6 h — the clearest intervention target.
- Platform choice (TikTok vs Instagram Reels vs YouTube Shorts) has no statistically meaningful effect on cognitive outcomes.

Key Recommendations

- Implement a 2–3 h/day usage cap as the evidence-based intervention threshold.
- Design age-specific digital wellness programmes targeting the 19–24 segment.
- Introduce sleep-awareness nudges into platform UX for users identified as sleep-deprived.

3. Sector and Business Context

Sector Overview

The short-form video segment has grown to represent the dominant mode of social media engagement globally. TikTok alone surpassed 1.5 billion monthly active users by 2024, while Meta's Instagram Reels and Google's YouTube Shorts compete aggressively for the same attention economy. The sector sits at the intersection of consumer technology, digital health, and behavioural psychology.

Decision-Maker / Stakeholder

Stakeholder	Interest in This Analysis
Platform Product Teams	Usage design, recommendation algorithm tuning, wellness features
Digital Health NGOs / Regulators	Evidence base for screen-time regulation policy
Parents & Educators	Identifying at-risk youth segments for intervention
Mental Health Researchers	Longitudinal study design and cohort targeting

Why This Problem Matters

Attention span is a foundational cognitive resource. Degradation of sustained attention correlates with reduced academic performance, lower occupational productivity, and increased mental health burden. Cross-sectional studies suggest heavy short-form content users demonstrate measurable reductions in task-switching ability and working memory capacity. With 48.2% of this dataset's users already exceeding what behavioural science considers safe daily consumption, the public health stakes are significant.

4. Problem Statement and Objectives

Formal Problem Definition

Research Question: How might we analyse short-form content consumption patterns to understand their impact on attention span across different user segments?

Objectives

O1: Characterise the distribution of daily short-form content consumption across the surveyed population.

O2: Identify demographic and behavioural segments with the greatest cognitive performance deficit.

O3: Determine whether platform choice, stress level, or sleep deprivation independently or jointly predict cognitive outcomes.

O4: Propose evidence-based thresholds for safe daily consumption that could inform product design or policy.

O5: Deliver actionable, dashboard-ready insights for non-technical stakeholders.

Scope

This study is cross-sectional, based on a single survey wave of 8,926 users aged 15–44. Only TikTok, Instagram Reels, and YouTube Shorts are in scope. Causality cannot be established from this design; findings are associative and directional. Generalisation to populations outside the 15–44 age band should be done with caution.

Success Criteria

Analysis will be considered successful if it can: (a) identify at least one statistically distinguishable high-risk user segment, (b) produce a defensible consumption threshold, and (c) deliver findings in a Tableau dashboard accessible to non-technical decision-makers.

5. Data Description

Source

Dataset: *Kaggle Link* - dataset compiled for academic research purposes as part of the DVA Course Project. The dataset was accessed via the project's shared data repository and loaded directly into the cleaning notebook.

Dataset Size and Coverage

8,926	8,779	10	3
Raw Records	Clean Records (for EDA)	Variables	Platforms Covered

Key Columns

Column	Type	Description	Range / Values
user_id	String	Unique user identifier	—
age	Numeric	User age in years	15 – 44
reels_watch_time_hours	Numeric	Daily reels consumption	0.5 – 6.0 h
daily_screen_time_hours	Numeric	Total daily screen time	1.0 – 13+ h
sleep_hours	Numeric	Average nightly sleep	3.0 – 10.0 h
attention_span_score	Numeric	Self-rated attention (1–10)	1 – 10
focus_level	Numeric	Self-rated focus (1–10)	1 – 10
task_completion_rate	Numeric	% tasks completed	38 – 100%
stress_level	Categorical	Perceived stress tier	Low / Medium / High
platform	Categorical	Primary platform used	TikTok / Instagram Reels / YouTube Shorts

Data Quality Issues Identified

The raw dataset contained multiple quality issues including inconsistent platform labels ('IG', 'insta' for Instagram; 'yt', 'youtube' for YouTube), numeric columns stored as strings, values violating domain constraints (e.g. age outside 10–100), missing values, and duplicate records. Full remediation is described in Section 6.

6. Cleaning and Transformation

All cleaning was performed in Python using pandas and NumPy in *01_Extraction_Cleaning.ipynb*. The pipeline proceeded in eight sequential stages:

1. Data Type Correction

Seven numeric columns (age, reels_watch_time_hours, daily_screen_time_hours, sleep_hours, attention_span_score, focus_level, task_completion_rate) were cast from object to float64 using pd.to_numeric with errors='coerce'. Invalid non-parseable values were set to NaN for downstream imputation.

2. Categorical Standardisation

String columns were stripped of leading/trailing whitespace and title-cased. Platform labels were mapped to canonical names: 'IG'/'insta' → 'Instagram Reels'; 'yt'/'youtube' → 'YouTube Shorts'; 'tik tok' → 'TikTok'. Unrecognised values were labelled 'Unknown' and later excluded from analysis.

3. Domain Rule Enforcement

Real-world constraints were applied: age in [10, 100]; all time columns ≤ 24 hours; reels_watch_time_hours ≤ daily_screen_time_hours; attention and focus scores in [1, 10]; task_completion_rate in [0, 100]. Rows violating any constraint were removed.

4. Missing Value Treatment

After coercion, remaining NaN values in numeric columns were filled using median imputation (robust to skewed distributions). Categorical columns received 'Unknown' as the fill value.

5. Outlier Handling

IQR-based clipping was applied to reels_watch_time_hours, daily_screen_time_hours, and sleep_hours. Clipping (not removal) was chosen to preserve the realistic behaviour of extreme users while preventing distributional distortion.

6. Logical Consistency Checks

A final consistency pass enforced the behavioural constraint that reels watch time cannot exceed total screen time. Any residual violations were removed.

7. Duplicate Removal

Exact duplicate rows were identified via df.duplicated() and dropped.

8. Final Validation

Post-cleaning: zero missing values, all columns in correct types, distributions within expected ranges, and categorical fields standardised. Output saved as *cleaned_data.csv*.

Output Dataset

The cleaned dataset contains 8,926 records with 10 columns. A further 147 'Unknown' records in stress_level/platform were excluded from comparative analysis, leaving 8,779 usable records for EDA — a retention rate of 98.4%.

7. KPI Framework

KPI	Formula	Purpose
Cognitive Performance Index (CPI)	(attention/10 + focus/10 + task_rate/100) / 3 × 10	Single composite score normalising three cognitive outcome variables into a 0–10 scale; primary outcome KPI.
% Over-Consuming Users	COUNT(reels > 3h) / COUNT(all) × 100	Operationalises the 3-hour threshold as the boundary between moderate and heavy use.
Reels Share of Screen Time (%)	reels_watch_time_hours / daily_screen_time_hours × 100	Measures how much of total screen time is dedicated to short-form content; useful for platform-level benchmarking.
Danger Zone User %	COUNT(reels>3h AND sleep<6h) / COUNT(all) × 100	Identifies compound-risk users — the primary intervention target.
Risk Segment	IF reels>3 AND sleep<6 → Danger Zone IF reels>3 ONLY → Reels at Risk IF sleep<6 ONLY → Sleep Deprived ELSE → Safe Zone	Four-way behavioural segmentation enabling targeted policy design.

8. Exploratory Analysis

Univariate Findings

All seven numeric variables were examined individually. Notably, attention_span_score, focus_level, and task_completion_rate display near-uniform (platykurtic) distributions with negative kurtosis values between −0.2 and −1.2. This flat shape — scores spread almost equally across the full 1–10 range — indicates high individual variability and mean group means will hover near the midpoint (~5.5) regardless of consumption levels, making difference detection harder.

Variable	Mean	Std Dev	Skewness	Kurtosis	Key Observation
Age	29.6	8.67	~0.0	−1.2	Near-uniform; slight older-adult bias
Reels Watch Time (h)	3.0	1.54	+0.18	−0.9	Right-skewed; heavy tail of 4–6h users
Daily Screen Time (h)	7.7	2.8	+0.12	−0.7	115 extreme outliers above 13h
Sleep Hours	6.5	1.5	−0.08	−0.6	Bimodal; 40.2% below 6h
Attention Span Score	5.50	2.59	~0.01	−1.19	Uniform — no dominant score cluster
Focus Level	5.51	2.59	~0.01	−1.19	Uniform — mirrors attention score pattern
Task Completion Rate	69.1	17.17	−0.03	−1.16	Uniform across 38–100% range

Bivariate Findings

Pearson correlations between all behavioral inputs (reels time, screen time, sleep) and all cognitive outputs (attention, focus, task completion, CPI) are extremely close to zero — maximum |r| = 0.221 (reels vs screen time, expected since reels is a subset). All other pairs fall below |r| = 0.03. Scatter plots with regression lines confirm flat slopes throughout.

The most informative bivariate finding is the **Age Group vs Attention Score** relationship: young adults aged 19–24 score the lowest (mean 5.38) while the 37–44 group scores highest (5.62). The 19–24 group also consumes the most reels per day (3.1 h), making them both the heaviest consumers and the most cognitively vulnerable — a critical alignment.

ANOVA testing across stress level groups and platform groups confirmed no statistically significant difference in any cognitive outcome variable. Mean attention scores across Low/Medium/High stress were 5.44, 5.54, and 5.54 respectively — within the noise band.

Multivariate Findings

Multivariate analysis was essential to surface the patterns invisible in pairwise comparisons. A heatmap of **Age Group × Reels Consumption Tier → Mean Attention Score** revealed the study's most actionable finding: the 19–24 age group combined with 2–4 h daily consumption registers the lowest attention scores in the entire matrix (5.27 and 5.33), while the 37–44 group at equivalent consumption levels remains above

5.65. This identifies *age as a moderating variable*.

Risk segmentation classified all users into four groups based on two binary rules (reels > 3h AND sleep < 6h):

Segment	% of Users	Approx. Count	CPI
Safe Zone	31.1%	2,776	5.98
Reels at Risk	28.8%	2,571	6.01
Sleep Deprived	20.8%	1,856	6.00
Danger Zone	19.3%	1,723	5.99

The ECDF analysis showed that a 2-hour daily cap would require behavioral change from more than 50% of the surveyed population — underscoring the depth of the over-consumption problem.

9. Statistical Analysis

Method	Variables	Result	Business Interpretation
Pearson Correlation	All numeric pairs	Max $\|r\| = 0.221$ (reels vs screen time); all cognitive pairs < 0.03	No linear predictor of attention — multi-factor model needed
IQR Outlier Detection	7 numeric columns	Only daily_screen_time has outliers (115 records, 1.31%)	Data is largely well-bounded; extreme users are a small but real minority
One-Way ANOVA	Stress level vs attention/focus/CPI	$p > 0.05$ for all comparisons; no significant group difference	Stress level alone cannot explain cognitive performance variation
One-Way ANOVA	Platform vs all cognitive outcomes	$p > 0.05$ for all comparisons	Platform choice is irrelevant to cognitive outcomes; behaviour matters, not app choice
Skewness / Kurtosis	attention, focus, task_completion	Skewness ≈ 0.0; Kurtosis ≈ −1.19 (uniform distribution)	Scores are genuinely individually variable — population-level effects are masked
Bivariate Pivot	Age group × Reels bin → Attention	19–24 + 2–4h → lowest scores (5.27–5.33)	Age moderates the consumption-cognition relationship; young adults are most susceptible

10. Dashboard Walkthrough

Four Tableau dashboards were designed to convert the EDA findings into operationally useful views. All dashboards share a consistent colour grammar: Teal (#1ABC9C) = safe/good, Amber (#F39C12) = caution, Red (#C0392B) = danger. Three global filters (Platform, Age Group, Stress Level) apply across all worksheets.

Dashboard 1 — Consumption Exposure

Answers: How much are people actually consuming?

- KPI cards: Avg Reels/Day (3.0h), Total Users (8,926), % Watching 3h+ (48.2%), Avg Screen Time (7.7h)
- Histogram of reels watch time with 3h over-consumption reference line
- Colour-coded bar chart by consumption tier (green → red scale)
- Average screen time by platform (TikTok, Instagram Reels, YouTube Shorts)
- Donut chart showing equal platform split (~33% each)

Dashboard 2 — Cognitive Health Overview

Answers: What is the overall state of attention and focus?

- KPI cards: Avg Attention (5.50), Avg Focus (5.51), Avg Task Completion (69.1%), Avg CPI
- Uniform distribution histograms for attention and focus — annotated as a key finding
- Age group bar chart: 19–24 highlighted red as most vulnerable segment
- Scatter plot: Reels vs Attention with linear trend line confirming near-zero R^2
- Task completion by stress level — no significant variation across groups

Dashboard 3 — Risk Segmentation

Answers: Who is most at risk and why?

- Hero donut chart: four-segment risk classification (Safe Zone 31.1%, Danger Zone 19.3%)
- Heatmap: Age Group × Reels Bin → Mean Attention (red-green diverging, centre = 5.5)
- Stacked bar: Stress composition by platform (confirming platform-independence)
- Danger Zone breakdown by age group: 19–24 represent the largest Danger Zone cohort
- Risk KPI cards: % Danger Zone, % Safe Zone, Avg Sleep of At-Risk users

Dashboard 4 — Intervention Threshold

Answers: Where should we draw the line?

- CPI bar chart by reels consumption tier with Y-axis range 5.0–6.5 (to surface subtle differences)
- Box plot: Attention score distribution by tier — confirms wide spread at all consumption levels
- Sleep by consumption tier with 6h sleep-risk reference line
- % users exceeding each threshold (2h cap affects 50%+ of users)
- Recommendation KPI: Evidence-based threshold of 2–3 hours/day

11. Key Insights

No single variable predicts cognitive decline.

All individual Pearson correlations between consumption/sleep variables and attention/focus/CPI are below |r| = 0.03. Cognitive outcomes are multi-factorial — single-variable interventions will be insufficient.

Young adults (19–24) are the most vulnerable segment.

The 19–24 age group has the lowest mean attention score (5.38) and the highest average reels consumption (3.1 h/day). Both conditions are jointly present, making this the highest-priority intervention cohort.

19.3% of users are in the Danger Zone.

Approximately 1,723 users simultaneously exceed 3 h daily reels AND sleep under 6 hours per night. This compound-risk group represents the clearest target for immediate platform-side wellness interventions.

Platform choice does not affect cognitive outcomes.

TikTok, Instagram Reels, and YouTube Shorts produce statistically indistinguishable results across all cognitive and behavioral variables. The problem is consumption volume and sleep, not the specific application.

Age moderates the consumption-cognition relationship.

The Age × Reels consumption heatmap reveals that 19–24 users at moderate consumption (2–4h) score worse than 37–44 users at high consumption (4–6h). Age acts as a protective factor in older adults — possibly due to more established attentional habits.

48.2% of users are over-consuming.

Nearly half the sample watches 3+ hours of reels daily. A 2-hour cap — the lower end of the evidence-based safe range — would require behavioral change from more than 50% of the population.

Sleep and stress interact non-linearly.

Medium stress + 6–7h sleep produces the highest CPI (6.11), while High stress + 7h+ produces the lowest (5.94). More sleep is not universally beneficial — the stress context modulates its cognitive value.

Cognitive score distributions are uniform, not normal.

Attention, focus, and task completion are spread nearly flat from 1–10. This indicates strong individual-level variation that averages out in group comparisons — a critical methodological note for any intervention effect study.

The data captures adaptation, not damage.

Near-zero correlations in this snapshot study do not mean content is harmless. Users have adapted to their consumption habits. Longitudinal designs consistently show attention degradation — the cross-sectional design masks cumulative effects.

40.2% of users sleep under 6 hours nightly.

Well below the 8-hour medical recommendation. Sleep deprivation is a co-occurring risk that interacts with content consumption — and is itself a significant independent health concern.

12. Recommendations

1) Implement a 2–3 Hour Daily Usage Cap

The ECDF analysis shows that a 2-hour cap affects 50%+ of users, while 3 hours is the current consensus threshold for over-consumption. Platform UX should introduce progressive usage friction after 2 hours (e.g. session summaries, pause prompts) and configurable hard limits at 3 hours.

Expected Impact: Estimated reduction of Danger Zone population from 19.3% to under 10% if adherence reaches 60%.

2) Deploy Age-Segmented Digital Wellness Programmes

Given that 19–24 year olds are both the heaviest consumers and the most cognitively vulnerable, targeted interventions should prioritise this cohort. University partnerships, in-app awareness modules, and attention-span training tools should be designed for this demographic.

Expected Impact: Potential 0.15–0.25 point improvement in mean attention score for the 19–24 segment.

3) Integrate Sleep-Awareness Nudges into Platform UX

Since 40.2% of users sleep under 6 hours, platforms should introduce opt-in bedtime reminders, night-mode auto-activation after 10 PM, and sleep-linked usage reports. This addresses the co-occurring risk factor independent of reels volume.

Expected Impact: Reduction of Sleep Deprived segment from 20.8% toward the population norm of <15%.

4) Conduct Longitudinal Follow-Up Study

The cross-sectional design cannot establish causality. A 12-month panel study tracking attention scores before and after consumption habit changes would provide the causal evidence needed to make regulatory-grade recommendations.

Expected Impact: Enables transition from associative to causal evidence base; supports policy advocacy.

5) Standardise CPI as a Platform Wellness Benchmark

The Cognitive Performance Index (CPI) should be adopted as an industry-standard metric, enabling cross-platform, cross-study comparison of wellness outcomes. Platforms could report quarterly CPI trends as part of digital wellbeing transparency reports.

Expected Impact: Provides a common measurement language for regulators, researchers, and product teams.

13. Limitations and Next Steps

Data Limitations

- **Cross-sectional design:** A single survey wave captures a snapshot, not a trajectory. Before/after comparisons are impossible, and causality cannot be established.
- **Self-reported measures:** Attention score, focus level, stress level, and sleep hours are all subjective self-ratings, subject to individual interpretation bias and social desirability effects.
- **Balanced categorical groups:** The near-perfect 33/33/33 split across platforms and stress levels is atypical for real-world data and may indicate stratified sampling. Generalisability to organic populations should be treated with caution.
- **Unknown records:** 147 records marked 'Unknown' in platform or stress_level were excluded, representing 1.65% of the dataset.

Method Limitations

- **Uniform score distributions:** The platykurtic (near-uniform) distributions of cognitive scores limit the sensitivity of correlation and ANOVA analyses — effect sizes that exist may be statistically invisible.
- **No temporal sequencing:** The dataset provides no information about when consumption habits were established, making it impossible to distinguish acute from chronic effects.
- **Linearity assumption:** Pearson correlation only captures linear relationships. Non-linear effects (e.g. threshold effects below which consumption is harmless) may exist but were not modelled.

Suggested Future Work

Initiative	Rationale	Timeline
12-month longitudinal panel study	Establish causal direction of consumption → attention decline	12–18 months
Objective attention measurement	Replace self-rated scores with neuropsychological tests (e.g. Stroop, sustained attention tasks)	6–12 months
Algorithm exposure analysis	Analyse whether recommendation algorithm aggressiveness (binge-triggering) is a mediating factor	6 months
Intervention RCT	Randomised trial of platform-side caps (2h vs 3h vs control) measuring CPI over 90 days	9–12 months

14. Contribution Matrix

The table below reflects the team's division of work across the project. GitHub commit history and pull request records should be referenced to validate the contributions below.

Contribution Matrix

This table must match evidence in GitHub Insights, PR history, and committed files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Neelanshu Karn			✓	✓			✓
Hrishi Dongre	✓		✓		✓		
Akhilesh Kumar	✓				✓	✓	
Lakshya Agrawal	✓			✓	✓		
Kapil Kumar Mathur			✓	✓		✓	
Ritik Atri	✓					✓	✓
Aaryan Gera		✓	✓	✓			

Note: This section should be updated with actual GitHub commit counts, PR links, and file paths before final submission. The contribution matrix will be cross-checked against the repository history during evaluation.